DIFFERENTIAL GEOMETRY AND LIE GROUPS

Jishnu Biswas, notes by Ramdas Singh

Fifth Semester

Contents

Chapter 1

PRELIMINARIES AND MANIFOLDS

July 20th.

The primary reference for this course will be Loring Tu's *An Introduction to Manifolds*. Prerequisites include Analysis of Several Variables, Linear Algebra, and Topology. The mid-term will account for 35% of the final grade, quizzes and student lectures will account for 15%, and the final exam will account for 50%.

Simply put, a manifold is any object that is locally Euclidean.

1.1 Calculus on $\mathbb{R}^n$

A point p or element in $\mathbb{R}^n$ is associated with an n-tuple of coordinates $x^1, x^2, \ldots, x^n$. Let U be an open neighbourhood of p in $\mathbb{R}^n$. For a non-negative integer k, we say that a function $f : U \to \mathbb{R}$ is C^k-*continuous* at $p \in U$ if all partial derivatives of f of order $\leq k$ exist and are continuous at p. f is said to be C^∞ or *smooth* if f is C^k for all $k \geq 0$. f is C^k (or C^∞)on U if it is C^k (or C^∞) at every point in U.

If f happens to be a vector-valued function, i.e. $f : U \to \mathbb{R}^m$, we say $f = (f^1, f^2, \ldots, f^m)$ is C^k (or C^∞) at p if each component function $f^i : U \to \mathbb{R}$ is C^k (or C^∞) at p. f is C^k (or C^∞) on U if it is C^k (or C^∞) at every point in U.

Definition 1.1. Let R be a commutative ring with unity. We term A to be an R-*algebra* if A is a commutative ring with unity and an R-module, and for $r \in R$ and $a, b \in A$, we have $r(ab) = (ra)b = a(rb)$.

In this course, we will be primarily be concerned with the case when R is a field, and A is an R-vector space.

For a subset $U \subseteq \mathbb{R}^n$, we denote $C^k(U)$ (or $C^\infty(U)$) as the set of all C^k (or C^∞) functions from U to $\mathbb{R}$. Note that this is a commutative ring with unity. It is also a vector space over $\mathbb{R}$. Together, $C^k(U)$ (or $C^\infty(U)$) is a (commutative) algebra over $\mathbb{R}$. Moreover, we have the descending chain of subalgebras

$$C^0(U) \supseteq C^1(U) \supseteq C^2(U) \supseteq \cdots C^k(U) \supseteq C^{k+1}(U) \supseteq \cdots \tag{1.1}$$

with

$$\bigcap_{k \geq 0} C^k(U) = C^\infty(U). \tag{1.2}$$

$f : U \to \mathbb{R}$ is termed *real analytic* at $p \in U$ if there exists a neighbourhood $V \subseteq U$ of p such that

$$f(x) = f(p) + \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}(p)(x^i - p^i) + \cdots + \frac{1}{k!} \sum_{1 \leq i_1, \ldots, i_k \leq n} \frac{\partial^k f}{\partial x^{i_1} \cdots \partial x^{i_k}}(p)(x^{i_1} - p^{i_1}) \cdots (x^{i_k} - p^{i_k}) + \cdots \tag{1.3}$$

for all $x \in V$. f is said to be real analytic on U if it is real analytic at every point in U. We denote the set of all real analytic functions on U as $C^\omega(U)$. Note that $C^\omega(U) \subseteq C^\infty(U)$.

Theorem 1.2 (*Taylor's theorem*). *Let $p \in U \subseteq \mathbb{R}^n$, and $f : U \to \mathbb{R}$ be a C^∞ function. Further assume that for all $x \in U$, the line segment from p to x is contained in U (U in this case is said to be* star-shaped *with respect to p). Then there exist $g_1, \ldots, g_n \in C^\infty(U)$ such that, for all $x \in U$,*

$$f(x) = f(p) + \sum_{i=1}^{n} g_i(x)(x^i - p^i), \tag{1.4}$$

with

$$g_i(p) = \frac{\partial f}{\partial x^i}(p), \quad 1 \leq i \leq n. \tag{1.5}$$

Proof. Let $x \in U$. Then the line segment $\{p + t(x-p) \mid t \in [0,1]\}$ is contained in U. Composing f with the map $t \mapsto p + t(x-p)$, we obtain

$$\frac{df}{dt}(p + t(x-p)) = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}(p + t(x-p))(x^i - p^i). \tag{1.6}$$

Integrating from 0 to 1, we get

$$f(x) - f(p) = \int_0^1 \frac{df}{dt}(p + t(x-p))dt = \sum_{i=1}^{n}(x^i - p^i) \int_0^1 \frac{\partial f}{\partial x^i}(p + t(x-p))dt = \sum_{i=1}^{n}(x^i - p^i)g_i(x). \tag{1.7}$$

Clearly, because f is C^∞, each g_i is also C^∞. Further, we have $g_i(p) = \int_0^1 \frac{\partial f}{\partial x^i}(p)dt = \frac{\partial f}{\partial x^i}(p)$. ■

Let $p = (p^1, \ldots, p^n), v \in \mathbb{R}^n$, and $f : U \to \mathbb{R}$ be C^∞ at p. The *directional derivative* of$f$ at p in the direction of v is defined as

$$\lim_{t \to 0} \frac{f(p + tv) - f(p)}{t} = \frac{d}{dt} f(p + tv)\Big|_{t=0} = \sum_{i=1}^{n} \frac{df}{dx^i}(p)v^i. \tag{1.8}$$

From here, we obtain the operator

$$D_v = \sum_{i=1}^{n} v^i \frac{\partial}{\partial x^i}\Big|_p \tag{1.9}$$

Now consider all pairs of the form (f, U) where $p \in U \subseteq \mathbb{R}^n$ is an open neighbourhood of p and $f : U \to \mathbb{R}$ is C^∞. Define an equivalence relation $\sim$ on this set by declaring $(f, U) \sim (g, V)$ if and only if there exists an open neighbourhood $W \subseteq U \cap V$ of p such that $f|_W = g|_W$. The equivalence class of (f, U) is denoted by $[(f, U)]$, and the set of all such equivalence classes is denoted by $C_p^\infty(\mathbb{R}^n)$. Such an equivalence class $[(f, U)]$ is called the *germ* of f at p. We can give $C_p^\infty(\mathbb{R}^n)$ the structure of a commutative ring with unity by defining

$$[(f, U)] + [(g, V)] = [(f + g, U \cap V)], \quad [(f, U)] \cdot [(g, V)] = [(fg, U \cap V)], \quad 1 = [(1, \mathbb{R}^n)]. \tag{1.10}$$

It can also be given the structure of a vector space over $\mathbb{R}$ by defining

$$\lambda[(f, U)] = [(\lambda f, U)], \quad \lambda \in \mathbb{R}. \tag{1.11}$$

Thus, $C_p^\infty(\mathbb{R}^n)$ is an $\mathbb{R}$-algebra. Now, given a germ $\alpha = [(f, U)] \in C_p^\infty(\mathbb{R}^n)$ and a vector $v \in \mathbb{R}^n$, we define

$$D_v \alpha = \sum_{i=1}^{n} v^i \frac{\partial f}{\partial x^i}\Big|_p = v \cdot \nabla f(p). \tag{1.12}$$

Note that the choice of f is immaterial, since if $[(f, U)] = [(g, V)]$, then there exists an open neighbourhood $W \subseteq U \cap V$ of p such that $f|_W = g|_W$, and hence $\nabla f(p) = \nabla g(p)$. One can infer that $D_{\lambda v}(f) = \lambda D_v(f) =$

$D_v(\lambda f)$ and $D_v(f+g) = D_v(f) + D_v(g)$ for all $\lambda \in \mathbb{R}$ and $f, g \in C_p^\infty(\mathbb{R}^n)$. Thus, D_v is an $\mathbb{R}$-linear map from $C_p^\infty(\mathbb{R}^n)$ to $\mathbb{R}$ (hereforth, we shall use f to also denote the germ $[(f,U)]$ of f at p). Also,

$$D_v(fg) = \sum_{i=1}^n v^i \frac{\partial (fg)}{\partial x^i}(p) = \sum_{i=1}^n v^i \left(f(p)\frac{\partial g}{\partial x^i}(p) + g(p)\frac{\partial f}{\partial x^i}(p) \right) = f(p)D_v(g) + g(p)D_v(f). \tag{1.13}$$

Such a property makes D_v what we call a derivation.

Definition 1.3. Any $\mathbb{R}$-linear map $T : C_p^\infty(\mathbb{R}^n) \to \mathbb{R}$ satisfying the Leibniz rule $T(fg) = f(p)T(g) + g(p)T(f)$ for all $f, g \in C_p^\infty(\mathbb{R}^n)$ is called a *derivation* at p.

One may show that if T is a derivation at p, then $T(1) = 0$.

The set of all derivations at p is denoted by $\mathfrak{D}_p(\mathbb{R}^n)$. It is easy to see that $\mathfrak{D}_p(\mathbb{R}^n)$ is a vector space over $\mathbb{R}$. Now consider the map

$$\theta : \mathbb{R}^n \to \mathfrak{D}_p(\mathbb{R}^n), \quad v \mapsto D_v. \tag{1.14}$$

This θ is an $\mathbb{R}$-linear map (verify). We claim that θ is an isomorphism of real vector spaces. To this end, let $v \in \mathbb{R}^n$ such that $D_v = 0 \in \mathfrak{D}_p(\mathbb{R}^n)$. Then $D_v f = 0$ for all $f \in C_p^\infty(\mathbb{R}^n)$. In particular, for the coordinate functions x^i, we have $D_v x^i = v_i = 0$ for all $1 \le i \le n$. Thus, $v = 0$, and hence θ is injective. To show surjectivity, we appeal to Taylor's theorem; let $D \in \mathfrak{D}_p(\mathbb{R}^n)$. We claim that $D = D_v$ where $v = (D(x^1), \dots, D(x^n))$. Indeed, for any $f \in C_p^\infty(\mathbb{R}^n)$, we have

$$\begin{aligned} D(f) &= D(f(p)) + D\left(\sum_{i=1}^n (x^i - p^i)g_i(x)\right) = 0 + \sum_{i=1}^n \big((x^i - p^i)(p)Dg_i(x) + D(x^i - p^i)g_i(p)\big) \\ &= \sum_{i=1}^n (Dx^i)g_i(p) = \sum_{i=1}^n \frac{\partial f}{\partial x^i}(p)Dx^i = D_v(f). \end{aligned} \tag{1.15}$$

Hence, θ is surjective, and we have the desired isomorphism $\mathfrak{D}_p(\mathbb{R}^n) \cong \mathbb{R}^n$.

Note that a germ $f \in C_p^\infty(\mathbb{R}^n)$ has a valid value at p, but no other point. Hence, we have the evaluation map

$$\mathrm{ev}_p : C_p^\infty(\mathbb{R}^n) \to \mathbb{R}, \quad [(f,U)] \mapsto f(p). \tag{1.16}$$

The kernel here is $\{[(f,U)] \in C_p^\infty(\mathbb{R}^n) \mid f(p) = 0\}$, which is a ideal in $C_p^\infty(\mathbb{R}^n)$. For any other element $[(f,U)] \in C_p^\infty(\mathbb{R}^n)$ with $f(p) \ne 0$, a valid inverse is given by $[(f^{-1}, U \setminus \{f = 0\})]$. This makes $C_p^\infty(\mathbb{R}^n)$ a local ring.

1.2 Multilinear Algebra

July 22nd.

Hereforth, V shall denote a finite-dimensional real vector space, and V^V shall denote its *dual space* $\mathrm{Hom}_{\mathbb{R}}(V, \mathbb{R})$. If $e_1, \dots, e_n$ are a basis for V, then the *dual basis* of V^V is denoted by $\alpha^1, \dots, \alpha^n \in V^V$, and is defined by $\alpha^i(e_j) = \delta_{i,j}$ for all $1 \le i, j \le n$. One may verify that the dual basis $\{\alpha^1, \dots, \alpha^n\}$, induced by the basis $\{e_1, \dots, e_n\}$, is indeed a basis for V^V. S_k shall denote the permutation group on k symbols, satisfying $|S_k| = k!$. For any $\sigma \in S_k$, the sign of σ is denoted by $\mathrm{sgn}(\sigma)$, and is defined as $+1$ if σ is an even permutation, and -1 if σ is an odd permutation.

With notation out of the way, let $k \ge 1$ be an integer. We shall consider the k-tuple $V^k = V \times \cdots \times V$ (k times).

Definition 1.4. A mapping $f : V^k \to \mathbb{R}$ is said to be *k-linear* if f is linear in each of its k arguments.

A special case is when $k = 2$, in which case f is called a *bilinear form*. Such an $f : V \times V \to \mathbb{R}$ satisfies

$$f(\lambda v + \mu w, u) = \lambda f(v,u) + \mu f(w,u), \quad f(u, \lambda v + \mu w) = \lambda f(u,v) + \mu f(u,w). \tag{1.17}$$

A common example of a bilinear form is any valid inner product $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ on V. Another example is the determinant $\det : \mathbb{R}^n \to \mathbb{R}$, where the vectors are interpreted as the columns of a matrix. We can also write

$$L_k(V) = \{f : V^k \to \mathbb{R} \mid f \text{ is } k\text{-linear}\}. \tag{1.18}$$

It is easy to see that $L_k(V)$ is an $\mathbb{R}$-vector space.

Definition 1.5. A k-linear map $f : V^k \to \mathbb{R}$ is called a *symmetric k-linear form* if for all $\sigma \in S_k$,

$$f(v_1, \dots, v_k) = f(v_{\sigma(1)}, \dots, v_{\sigma(k)}). \tag{1.19}$$

That is, f is invariant under any permutation of its arguments. Moreover, f is called an *alternating k-linear form* if for all $\sigma \in S_k$,

$$f(v_1, \dots, v_k) = \operatorname{sgn}(\sigma) f(v_{\sigma(1)}, \dots, v_{\sigma(k)}). \tag{1.20}$$

One can see that the standard inner product on $\mathbb{R}^n$ is a symmetric bilinear form, and the determinant is an alternating n-linear form. Related to these definitions, we can immediately define the following subspaces of $L_k(V)$:

$$S_k(V) = \{f \in L_k(V) \mid f \text{ is symmetric}\} \subseteq L_k(V), \quad A_k(V) = \{f \in L_k(V) \mid f \text{ is alternating}\} \subseteq L_k(V). \tag{1.21}$$

Currently, our main interest of study is $A_k(V)$, the space of alternating k-linear forms on V. Elements of $A_k(V)$ are termed *alternating k-tensors*. Similarly, those of $S_k(V)$ and $L_k(V)$ are called *symmetric k-tensors* and *k-tensors*, respectively. For $k = 0$, we define $L_0(V) = S_0(V) = A_0(V) = \mathbb{R}$. For $k = 1$, we have $L_1(V) = S_1(V) = A_1(V) = V^V$. We can define the (left) action of S_k on $L_k(V)$ by

$$(\sigma \cdot f)(v_1, \dots, v_k) = f(v_{\sigma(1)}, \dots, v_{\sigma(k)}). \tag{1.22}$$

Thus, f is symmetric if $\sigma \cdot f = f$ for all $\sigma \in S_k$, and f is alternating if $\sigma \cdot f = \operatorname{sgn}(\sigma) f$ for all $\sigma \in S_k$.

Given an $f \in L_k(V)$, we can define the *symmetrization* $Sf \in S_k(V)$, of f, as

$$Sf = \sum_{\sigma \in S_k} \sigma \cdot f. \tag{1.23}$$

One may, indeed, check that Sf is symmetric. Similarly, we can define the *alternation* $Af \in A_k(V)$, of f, as

$$Af = \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma) \sigma \cdot f. \tag{1.24}$$

1.2.1 The Tensor and Wedge Products

July 24th.

For any $f \in L_k(V)$ and $g \in L_l(V)$, we can define the *tensor product* $f \otimes g \in L_{k+l}(V)$ as

$$(f \otimes g)(v_1, \dots, v_{k+l}) = f(v_1, \dots, v_k) g(v_{k+1}, \dots, v_{k+l}). \tag{1.25}$$

Verify that $f \otimes g$ is indeed $(k+l)$-linear. We are interested in the tensor product of alternating tensors, so let $f \in A_k(V)$ and $g \in A_l(V)$. Then, we can define the *wedge product* $f \wedge g \in A_{k+l}(V)$ as

$$f \wedge g = \frac{1}{k!l!} A(f \otimes g) = \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) \sigma \cdot (f \otimes g). \tag{1.26}$$

The associativity of the tensor product holds easily: for $f \in L_k(V)$, $g \in L_l(V)$, and $h \in L_m(V)$,

$$(f \otimes g) \otimes h = f \otimes (g \otimes h). \tag{1.27}$$

Proposition 1.6. *For $f \in A_k(V)$ and $g \in A_l(V)$, we have*

$$f \wedge g = (-1)^{kl} g \wedge f. \tag{1.28}$$

Proof. We shall show that $A(f \otimes g) = (-1)^{kl} A(g \otimes f)$. To this end, fix the permutation $\tau \in S_{k+l}$ defined by

$$\tau = \begin{pmatrix} 1 & 2 & \cdots & l & l+1 & l+2 & \cdots & l+k \\ k+1 & k+2 & \cdots & k+l & 1 & 2 & \cdots & k \end{pmatrix}. \tag{1.29}$$

We have

$$\begin{aligned} A(f \otimes g)(v_1, \ldots, v_{k+l}) &= \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}) g(v_{\sigma(k+1)}, \ldots, v_{\sigma(k+l)}) \\ &= \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) f(v_{\sigma\tau(l+1)}, \ldots, v_{\sigma\tau(l+k)}) g(v_{\sigma\tau(1)}, \ldots, v_{\sigma\tau(l)}) \\ &= \operatorname{sgn}(\tau) \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma\tau) f(v_{\sigma\tau(l+1)}, \ldots, v_{\sigma\tau(l+k)}) g(v_{\sigma\tau(1)}, \ldots, v_{\sigma\tau(l)}) \\ &= \operatorname{sgn}(\tau) A(g \otimes f)(v_1, \ldots, v_{k+l}). \end{aligned} \tag{1.30}$$

With $\operatorname{sgn}(\tau) = (-1)^{kl}$, the proof is complete. ∎

Corollary 1.7. *For $f \in A_k(V)$, we have $f \wedge f = 0$ if k is odd.*

Proof. Easy since $f \wedge f = (-1)^{k^2} f \wedge f = -f \wedge f$ if k is odd, so $f \wedge f = 0$. ∎

Lemma 1.8. *For $f \in L_k(V)$ and $g \in L_l(V)$, we have*

$$A(Af \otimes g) = k! A(f \otimes g), \quad A(f \otimes Ag) = l! A(f \otimes g). \tag{1.31}$$

Proof. We shall prove the first equality; the second follows similarly. We have

$$\begin{aligned} A(Af \otimes g) &= \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) \sigma \cdot \left(\sum_{\tau \in S_k} \operatorname{sgn}(\tau)(\tau \cdot f) \otimes g \right) = \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) \sigma \cdot \left(\sum_{\tau \in S_k} \operatorname{sgn}(\tau) \tau \cdot (f \otimes g) \right) \\ &= \sum_{\tau \in S_k} \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma\tau) \sigma\tau \cdot (f \otimes g) = k! \sum_{\alpha \in S_{k+l}} \operatorname{sgn}(\alpha) \alpha \cdot (f \otimes g) = k! A(f \otimes g). \end{aligned} \tag{1.32}$$

∎

Proposition 1.9. *The wedge product is associative, i.e. for $f \in A_k(V)$, $g \in A_l(V)$, and $h \in A_m(V)$, we have*

$$(f \wedge g) \wedge h = f \wedge (g \wedge h). \tag{1.33}$$

Proof. We have

$$\begin{aligned} (f \wedge g) \wedge h &= \frac{1}{(k+l)!m!} A((f \wedge g) \otimes h) = \frac{1}{(k+l)!m!k!l!} A(A(f \otimes g) \otimes h) = \frac{(k+l)!}{(k+l)!l!k!m!} A((f \otimes g) \otimes h) \\ &= \frac{1}{k!l!m!} A(f \otimes g \otimes h) = f \wedge (g \wedge h). \end{aligned} \tag{1.34}$$

∎

Corollary 1.10. *$f \wedge g \wedge h = \frac{1}{k!l!m!} A(f \otimes g \otimes h)$ for $f \in A_k(V)$, $g \in A_l(V)$, and $h \in A_m(V)$. Inductively,*

$$f_1 \wedge \cdots \wedge f_k = \frac{1}{k_1! \cdots k_k!} A(f_1 \otimes \cdots \otimes f_k), \tag{1.35}$$

where $f_i \in A_{k_i}(V)$ for $1 \leq i \leq k$.

Proposition 1.11. *Let $\alpha^1, \dots, \alpha^n \in V^V = A_1(V) = L_1(V)$, and $v_1, \dots, v_k \in V$. Then*

$$(\alpha^1 \wedge \cdots \wedge \alpha^k)(v_1, \dots, v_k) = \det(\alpha^i(v_j))_{1 \leq i,j \leq k}. \tag{1.36}$$

Proof. We have

$$\begin{aligned}(\alpha^1 \wedge \cdots \wedge \alpha^k)(v_1, \dots, v_k) &= \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma)(\alpha^1 \otimes \cdots \otimes \alpha^k)(v_{\sigma(1)}, \dots, v_{\sigma(k)}) \\ &= \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma) \alpha^1(v_{\sigma(1)}) \alpha^2(v\sigma(2)) \cdots \alpha^k(v_{\sigma(k)}) = \det(\alpha^i(v_j))_{1 \leq i,j \leq k}.\end{aligned} \tag{1.37}$$

■

July 27th.

Let V be an $\mathbb{R}$-vector space with basis $e_1, \dots, e_n$, and V^V be its dual space with dual basis $\alpha^1, \dots, \alpha^n$. Moreover, let I be a multi-index of length k, i.e. $I = (i_1, \dots, i_k)$ with $1 \leq i_1, \dots, i_k \leq n$. We define $e_I := (e_{i_1}, \dots, e_{i_k}) \in V^k$ and $\alpha^I := \alpha^{i_1} \wedge \cdots \wedge \alpha^{i_k} \in A_k(V)$. Then, a k-linear form $f \in L_k(V)$ is completely determined by its values on the e_I, and an alternating k-linear form $f \in A_k(V)$ is completely determined by its values on the e_I with $i_1 < i_2 < \cdots < i_k$.

To this end, we first claim that $\alpha^I(e_J) = \delta^I_J$ for all multi-indices I, J of length k. Indeed, if $J = (j_1, \dots, j_k)$ with $j_1 < j_2 < \cdots < j_k$, then

$$\alpha^I(e_J) = \det(\alpha^{i_m}(e_{j_n}))_{1 \leq m,n \leq k}; \tag{1.38}$$

. this is 1 for $I = J$. Suppose $I \neq J$. Let $1 \leq l \leq k$ be the first index where $i_l \neq j_l$. Then the l^{th} row of $\alpha^{i_m}(e_{j_n})_{1 \leq m,n \leq k}$ is an all zeroes row, and hence $\det(\alpha^{i_m}(e_{j_n}))_{1 \leq m,n \leq k} = 0$. Thus, $\alpha^I(e_J) = \delta^I_J$. We now claim that α^I, for strictly increasing multi-indices I of length k, form a basis for $A_k(V)$. For linear independence, we simply have (the sums in this context are over strictly increasing multi-indices I of length k):

$$\sum_I c_I \alpha^I = 0 \implies \sum_I c_I \alpha^I(e_J) = 0 \implies c_J = 0, \quad \forall J. \tag{1.39}$$

To show that they span $A_k(V)$, let $f \in A_k(V)$. Then consider the k-linear form

$$g = \sum_I f(e_I) \alpha^I. \tag{1.40}$$

We show that $f = g$; it is enough to prove that $f(e_J) = g(e_J)$ for all strictly increasing multi-indices J of length k. We have

$$g(e_J) = \sum_I f(e_I) \alpha^I(e_J) = f(e_J) \alpha^J(e_J) + \sum_{I \neq J} f(e_I) \alpha^I(e_J) = f(e_J). \tag{1.41}$$

Corollary 1.12. *$A_k(V) = 0$ if $k > n$. Further, $\dim A_k(V) = \binom{n}{k}$ if $1 \leq k \leq n$.*

1.3 Building Blocks for Manifolds

Recall that we had defined the class of germs $C^\infty_p(\mathbb{R}^n)$ of C^∞ functions at a point $p \in \mathbb{R}^n$, which agree on some neighbourhood of p. One can show that $C^\infty_p(\mathbb{R}^n)$ and $C^\infty_p(U)$ are isomorphic as $\mathbb{R}$-algebras, where $U \subseteq \mathbb{R}^n$ is an open neighbourhood of p. Note that we term $T_p(U) = T_p(\mathbb{R}^n) = \mathbb{R}^n$ as the *tangent space* of U (or $\mathbb{R}$) at p.

Definition 1.13. Let $U \subseteq \mathbb{R}^n$ be an open subset. A *vector field* on U is a function $X : U \to \amalg_{p\in U} T_p(\mathbb{R}^n)$ such that $X(p) \in T_p(\mathbb{R}^n)$ for all $p \in U$. In particular,

$$X(p) = \sum_{i=1}^{n} a^i(p) \frac{\partial}{\partial x^i}\Big|_p, \quad a^i(p) \in \mathbb{R}. \tag{1.42}$$

We can rewrite this as

$$X = \sum_{i=1}^{n} a^i \frac{\partial}{\partial x^i}, \quad a^i : U \to \mathbb{R}. \tag{1.43}$$

X is called a *smooth vector field* if each a^i is a smooth function on U. The set of all smooth vector fields on U is denoted by $\mathfrak{X}(U)$.

This is a real vector space; note that for $X, Y \in \mathfrak{X}(U)$, we have $X + Y \in \mathfrak{X}(U)$, and for $\lambda \in \mathbb{R}$, we have $\lambda X \in \mathfrak{X}(U)$. Further, $\mathfrak{X}(U)$ is a $C^\infty(U)$-module since for $f \in C^\infty(U)$ and $X \in \mathfrak{X}(U)$, we have $fX \in \mathfrak{X}(U)$, where

$$(fX)(p) = f(p)X(p), \quad p \in U. \tag{1.44}$$

Notice that $X \in \mathfrak{X}(U)$ describes a map $C^\infty(U) \to C^\infty(U)$, where $f \mapsto Xf$, being defined as $(Xf)(p) = X_pf$. This is a derivation on $C^\infty(U)$, since for $f, g \in C^\infty(U)$, we have

$$(X(fg))(p) = X_p(fg) = f(p)X_pg + g(p)X_pf = (fXg)(p) + (gXf)(p). \tag{1.45}$$

We refine our definition of a derivation as follows.

Definition 1.14. Let K be a field, and A be a K-algebra. A *derivation* $D : A \to A$ is an $\mathbb{R}$-linear map satisfying the Leibniz rule $D(ab) = aD(b) + bD(a)$ for all $a, b \in A$.

Definition 1.15. A *differential 1-form* on U is a smooth function $\omega : U \to \amalg_{p\in U} T_p(\mathbb{R}^n)^V$ such that $\omega(p) \in T_p(\mathbb{R}^n)^V =: T_p^*(\mathbb{R}^n)$ for all $p \in U$.

For example, for $f \in C^\infty(U)$, we can define the differential 1-form df on U by $df(p) = df_p \in \mathrm{Hom}_{\mathbb{R}}(T_p\mathbb{R}^n, \mathbb{R})$, where $df_p(X) = X_pf$ for all $X \in T_p\mathbb{R}^n$. In particular, dx^i is a differential 1-form on U for all $1 \le i \le n$. In fact, $\{dx^1, \dots, dx^n\}$ is the dual basis of $T_p^*\mathbb{R}^n$ corresponding to the basis $\{\frac{\partial}{\partial x^1}\Big|_p, \dots, \frac{\partial}{\partial x^n}\Big|_p\}$ of $T_p\mathbb{R}^n$. Thus, a differential 1-form ω on U can be expressed as

$$\omega(p) = \sum_{i=1}^{n} a_i(p) dx^i\Big|_p, \quad a_i : U \to \mathbb{R}. \tag{1.46}$$

$T_p^*(\mathbb{R}^n)$, here, is termed the *cotangent space* of $\mathbb{R}^n$ at p. Thus, our current setup is as follows: $V = T_p\mathbb{R}^n$, and $V^V = T_p^*\mathbb{R}^n$. The basis for $T_p\mathbb{R}^n$ considered is $\{\frac{\partial}{\partial x^1}\Big|_p, \dots, \frac{\partial}{\partial x^n}\Big|_p\}$, and the corresponding dual basis for $T_p^*\mathbb{R}^n$ is $\{dx^1\Big|_p, \dots, dx^n\Big|_p\}$. And finally, we have

$$A_k(T_p\mathbb{R}^n) = \text{the space of alternating } k\text{-linear forms on } T_p\mathbb{R}^n \tag{1.47}$$

which has the basis $\{dx^{i_1}\Big|_p \wedge \dots \wedge dx^{i_k}\Big|_p \mid 1 \le i_1 < \dots < i_k \le n\}$.

Definition 1.16. A *differential k-form* on U is a smooth function $\omega : U \to \amalg_{p\in U} A_k(T_p\mathbb{R}^n)$ such that $\omega(p) \in A_k(T_p\mathbb{R}^n)$ for all $p \in U$.

Any such ω may be written as $\omega = \sum_I a_I dx_I$. It termed a smooth differential k-form if each a_I is a smooth function on U. The set of all smooth differential 1-forms on U is denoted by $\Omega^1(U)$, and the set of all smooth differential k-forms on U is denoted by $\Omega^k(U)$. Both these sets are $\mathbb{R}$-vector spaces, and $\Omega^k(U)$ is a free $C^\infty(U)$-module of rank $\binom{n}{k}$. The isomorphism may be given as

$$\Omega^1(U) \cong C^\infty(U)^{\oplus \binom{n}{k}}, \quad \omega = \sum_I a_I dx_I \mapsto (a_1, a_2, \ldots, a_{\binom{n}{k}}). \tag{1.48}$$

We can endow $\Omega^k(U)$ with the wedge product $\wedge : \Omega^k(U) \times \Omega^l(U) \to \Omega^{k+l}(U)$, defined as

$$(\omega \wedge \eta)(p) = \omega(p) \wedge \eta(p), \quad p \in U, \tag{1.49}$$

or

$$\omega \wedge \eta = \sum_{I,J} a_I b_J dx_I \wedge dx_J \tag{1.50}$$

where I and J are strictly increasing multi-indices of length k and l, respectively. Note that if $I \cap J \neq \emptyset$, then $dx_I \wedge dx_J = 0$. Thus, the sum may be taken over all strictly increasing multi-indices I and J of length k and l, respectively, such that $I \cap J = \emptyset$. Also,

$$A_*(V) = \bigoplus_{k=0}^{n} A_k(V) \tag{1.51}$$

is defined as the *exterior algebra* of V. The wedge product on here is associative and anti-commutative. We define $\Omega^0(U) = C^\infty(U)$. The map $d : \Omega^k(U) \to \Omega^{k+1}(U)$ is defined as the *exterior derivative*; for $k = 0$, we have

$$d : C^\infty(U) \to \Omega^1(U), \quad f \mapsto df = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i} dx^i. \tag{1.52}$$

For a k-form $\omega = \sum_I a_I dx_I \in \Omega^k(U)$, we have

$$d\omega = \sum_I da_I \wedge dx_I = \sum_I \sum_{j=1}^{n} \frac{\partial a_I}{\partial x^j} dx^j \wedge dx_I. \tag{1.53}$$

Example 1.17. Let $U \subseteq \mathbb{R}^3$. Let us call the coordinates $x^1 = x$, $x^2 = y$, and $x^3 = z$. The only non-zero forms are $\Omega^0(U)$, $\Omega^1(U)$, $\Omega^2(U)$, and $\Omega^3(U)$. Note that $\Omega^0(U) = C^\infty(U)$, and there's nothing to say about this. For $\Omega^1(U)$, a form looks like $f\,dx + g\,dy + h\,dz$ for $f, g, h \in C^\infty(U)$. For $\Omega^2(U)$, a form looks like $R\,dx \wedge dy + Q\,dx \wedge dz + P\,dy \wedge dz$ for $f, g, h \in C^\infty(U)$.

July 29th.

The map $\Omega^1(U) \times \mathfrak{X}(U) \to C^\infty(U)$, defined as $(\omega, X) \mapsto \omega(X)$, is $\mathbb{R}$-bilinear. In particular, for $\omega = \sum_{i=1}^n a_i dx^i$ and $X = \sum_{j=1}^n b^j \frac{\partial}{\partial x^j}$, we have

$$\omega(X) = \sum_{i=1}^{n} a_i dx^i \left(\sum_{j=1}^{n} b^j \frac{\partial}{\partial x^j} \right) = \sum_{i=1}^{n} a_i b^i \in C^\infty(U). \tag{1.54}$$

Extending this, we have the map $\Omega^k(U) \times \mathfrak{X}(U)^k \to C^\infty(U)$, defined as $(\omega, X_1, \ldots, X_k) \mapsto \omega(X_1, \ldots, X_k)$, which is $\mathbb{R}$-multilinear. In particular, for $\omega = \sum_I b_I dx^I$ and $X_i = \sum_{i=1}^n a_i^j \frac{\partial}{\partial x^j}$, we have

$$\omega(X_1, \ldots, X_k) = \sum_I b_I a^1_{i_1} \cdots a^k_{i_k} \in C^\infty(U) \tag{1.55}$$

using the fact that $dx^I(\frac{\partial}{\partial x^{i_1}}, \ldots, \frac{\partial}{\partial x^{i_k}}) = 1$ if the multi-index matches, and 0 otherwise.

1.3.1 Exterior Differentiation

Here, d_k shall denote the map $d_k : \Omega^k(U) \to \Omega^{k+1}(U)$, defined as the *exterior derivative*. The subscript k is often omitted, and we simply write $d : \Omega^k(U) \to \Omega^{k+1}(U)$. For $k = 0$, we have

$$d : C^\infty(U) \to \Omega^1(U), \quad f \mapsto df = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i} dx^i. \tag{1.56}$$

Analogously, for a general k-form $\omega = \sum_I a_I dx^I \in \Omega^k(U)$, we have

$$d : \Omega^k(U) \to \Omega^{k+1}(U), \quad \omega \mapsto d\omega = \sum_I da_I \wedge dx^I = \sum_I \sum_{j=1}^n \frac{\partial a_I}{\partial x^j} dx^j \wedge dx^I. \tag{1.57}$$

All the d maps are $\mathbb{R}$-linear, but not $C^\infty(U)$-linear. Also, for two forms $\omega \in \Omega^k(U)$ and $\eta \in \Omega^l(U)$, we have

$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta. \tag{1.58}$$

This is known as *antiderivation*. Moreover, $d_{k+1} \circ d_k = d^2 = 0$. Both these properties may be verified by the reader. Now suppose we have another map $D : \Omega^k(U) \to \Omega^{k+1}(U)$ which is $\mathbb{R}$-linear, satisfies antiderivation, and $D^2 = 0$, and $Df = df$ for all $f \in C^\infty(U)$. Then $D = d$; the exterior derivative is unique. Thus, a chain of maps is formed:

$$0 \xrightarrow{d} C^\infty(U) \xrightarrow{d} \Omega^1(U) \xrightarrow{d} \Omega^2(U) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^n(U) \xrightarrow{d} 0 \quad \text{for open } U \subseteq \mathbb{R}^n. \tag{1.59}$$

An $\omega \in \Omega^k(U)$ is called an *exact k-form* if $\omega = d\eta$ for some $\eta \in \Omega^{k-1}(U)$. It is called a *closed k-form* if $d\omega = 0$. Note that every exact form is closed, but the converse is not true in general. We will refine the following later: quotienting the space of closed k-forms by the space of exact k-forms gives us the *de Rham cohomology* of U, denoted by $H^k_{\text{dR}}(U)$.

1.4 Manifolds

July 31st.

Definition 1.18. A topological space M is termed a *topological manifold* of dimension n if

1. M is Hausdorff,
2. M is secound countable (that is, has a countable basis), and
3. for all $p \in M$, there exists an open neighbourhood $U \subseteq M$ and a homemorphism $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^n$, where $\varphi(U)$ is open. This makes a manifold *locally euclidean*.

The *invariance of dimension* property states that if we have two open sets $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ such that U is homeomorphic to V, then $m = n$. Thus, the dimension of a topological manifold is well-defined.

Definition 1.19. A pair (U, φ) is called a *coordinate chart* on M if $U \subseteq M$ is open and $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^n$ is a homeomorphism.

A trivial example is $\mathbb{R}^n$ with one (coordinate) chart $(\mathbb{R}^n, \text{id}_{\mathbb{R}^n})$. In fact, if $U \subseteq \mathbb{R}^n$ is open, then (U, id_U) is a coordinate chart. Note that subspaces inherit the properties of Hausdorffness and second countability. We state some properties without proof; if X is locally Euclidean but not T_2, then X being a topological manifold implies that the number of connected (or path connected) components is at most countable.

Definition 1.20. Two charts (U, φ) and (V, ψ) on a topological manifold M are said to be C^∞-*compatible* if either

1. $U \cap V = \emptyset$, or
2. $U \cap V \neq \emptyset$, and the transition maps $\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$ and $\varphi \circ \psi^{-1} : \psi(U \cap V) \to \varphi(U \cap V)$ are C^∞-maps between open subsets of $\mathbb{R}^n$.

Similarly, C^k-compatibility may be defined for $k \geq 0$. A C^ω-manifold is termed to be a real analytic manifold. In case of $k = 2m$, then $\mathbb{R}^{2m}$ is identified with $\mathbb{C}^m$, and the transition maps are required to be holomorphic.

Definition 1.21. M is a C^∞-manifold or a *smooth manifold* if M is a topological manifold and there exists a family of charts $\{(U_\alpha, \varphi_\alpha)\}_{\alpha\in\Lambda}$ such that

1. $\{U_\alpha\}_{\alpha\in\Lambda}$ is an open cover of M, and
2. $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) are C^∞-compatible for all $\alpha, \beta \in \Lambda$.

Similarly, C^k-manifolds, C^ω-manifolds, and complex manifolds may be defined. The family of charts $\{(U_\alpha, \varphi_\alpha)\}_{\alpha\in\Lambda}$ is called an *atlas* of M.

The most basic non-trivial example is perhaps the circle $S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$, which is a C^∞-manifold of dimension 1. We can think of $\mathbb{R}^2$ as $\mathbb{C}$. Let $U = S^1 \setminus \{1\} = \{e^{it} \mid 0 < t < 2\pi\}$ and $V = S^1 \setminus \{-1\} = \{e^{it} \mid -\pi < t < \pi\}$. Then (U, φ) and (V, ψ) are charts, where $\varphi : U \to (0, 2\pi) \subseteq \mathbb{R}$ is defined as $e^{it} \mapsto t$ and $\psi : V \to (-\pi, \pi) \subseteq \mathbb{R}$ is defined as $e^{it} \mapsto t$. Here, $U \cap V = \{e^{it} \mid 0 < t < \pi\} \cup \{e^{-t} \mid \pi < t < 2\pi, -\pi < t < 0\}$. The transition maps are $\psi \circ \varphi^{-1} : (0, \pi) \to (0, \pi)$ defined as $t \mapsto t$ and $\psi \circ \varphi^{-1} : (\pi, 2\pi) \to (-\pi, 0)$ defined as $t \mapsto t - 2\pi$. Both these maps are smooth, and hence the charts are compatible. Thus, S^1 is a smooth manifold of dimension 1.

August 3rd.

Proposition 1.22. *Let $\mathcal{U} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha\in\Lambda}$ be an atlas on M. Suppose (V, ψ) and (W, σ) are two charts on M which are C^∞-compatible with all charts in $\mathcal{U}$. Then (V, ψ) and (W, σ) are C^∞-compatible.*

Proof. Let $p \in V \cap W$. Then there exists $\alpha \in \Lambda$ such that $p \in U_\alpha$. Since (V, ψ) and $(U_\alpha, \varphi_\alpha)$ are compatible, we have that $\psi \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap V) \to \psi(U_\alpha \cap V)$ is smooth. Similarly, since (W, σ) and $(U_\alpha, \varphi_\alpha)$ are compatible, we have that $\sigma \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap W) \to \sigma(U_\alpha \cap W)$ is smooth. Now, we have

$$(\sigma \circ \psi^{-1})|_{\psi(U_\alpha\cap V\cap W)} = (\sigma \circ \varphi_\alpha^{-1})|_{\varphi_\alpha(U_\alpha\cap V\cap W)} \circ (\varphi_\alpha \circ \psi^{-1})|_{\psi(U_\alpha\cap V\cap W)}. \tag{1.60}$$

The right-hand side is a composition of smooth maps, and hence is smooth. Thus, $\sigma \circ \psi^{-1}$ is smooth on an open neighbourhood of p. Since p was arbitrary, we have that $\sigma \circ \psi^{-1}$ is smooth on $V \cap W$. Similarly, we can show that $\psi \circ \sigma^{-1}$ is smooth on $V \cap W$. Thus, (V, ψ) and (W, σ) are compatible. ■

This leads to the following definition.

Definition 1.23. A *maximal atlas* $\mathcal{M}$ on a smooth manifold M is an atlas which is maximal with respect to inclusion; that is, if $\mathcal{M} \subseteq \mathcal{U}$ for some atlas $\mathcal{U}$, then $\mathcal{M} = \mathcal{U}$. Equivalently, $\mathcal{M}$ is a maximal atlas if it contains all charts which are compatible with all charts in $\mathcal{M}$.

Any atlas $\mathcal{U}$ on M is contained in a unique maximal atlas $\mathcal{M}$. For a proof of this, let $\mathcal{M} = \{(U, \varphi) \mid$ compatible with all charts in $\mathcal{U}\}$. Then $\mathcal{M}$ is an atlas. If $\mathcal{M} \subseteq \mathcal{M}'$, then any chart in $\mathcal{M}'$ must be compatible with all charts in $\mathcal{M}$; it must be compatible with all charts in $\mathcal{U}$. By definition of $\mathcal{M}$, such a chart must be in $\mathcal{M}$, so $\mathcal{M} = \mathcal{M}'$. For uniqueness, if $\mathcal{U} \subseteq \mathcal{M}, \mathcal{M}'$, then $\mathcal{M}, \mathcal{M}' \subseteq \mathcal{M} \cup \mathcal{M}'$ tells us $\mathcal{M} = \mathcal{M}'$ by maximality. The unique maximal atlas $\mathcal{M} \supseteq \mathcal{U}$ is also called a differentiable structure on M.

Example 1.24.

1. Let $U \subseteq \mathbb{R}^n$, and $F : U \to \mathbb{R}^m$ be a C^∞-map. The graph of F is defined as $\Gamma(F) = \{(x, F(x)) \mid x \in U\} \subseteq \mathbb{R}^{n+m}$. At atlas on $\Gamma(F)$ is given by $\{(U, \varphi)\}$, where $\varphi : U \to \Gamma(F)$ is defined as $x \mapsto (x, F(x))$. The transition map is $\varphi^{-1} : \Gamma(F) \to U$ defined as $(x, F(x)) \mapsto x$. This is smooth, and hence $\Gamma(F)$ is a smooth manifold of dimension n.
2. Let M and N be smooth manifolds of dimension m and n, respectively. Then $M \times N$ is a smooth manifold of dimension $m + n$; if $\mathcal{U}$ and $\mathcal{V}$ are atlases on M and N, respectively, then $\mathcal{U} \times \mathcal{V} = \{(U \times V, \varphi \times \psi) \mid (U, \varphi) \in \mathcal{U}, (V, \psi) \in \mathcal{V}\}$ is an atlas on $M \times N$. The transition maps are $(\varphi_2 \times \psi_2) \circ (\varphi_1 \times \psi_1)^{-1} = (\varphi_2 \circ \varphi_1^{-1}) \times (\psi_2 \circ \psi_1^{-1})$, which is smooth since both $\varphi_2 \circ \varphi_1^{-1}$ and $\psi_2 \circ \psi_1^{-1}$ are smooth.

1.4.1 Smooth Functions and Smooth Maps

We will often omit ‘smooth’ but still mean smooth in the context of manifolds.

Definition 1.25. Let M be a manifold of dimension n. A function $f : M \to \mathbb{R}$ is smooth at $p \in M$ if there exists a chart (U, φ) of M such that $f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}$ is smooth at $\varphi(p)$. A function $f : M \to \mathbb{R}$ is smooth if it is smooth at all $p \in M$.

The definition of the smooth function $f : M \to \mathbb{R}$ is independent of the choice of chart (U, φ), since if (V, ψ) is another chart containing p, then $\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$ is smooth, and hence $f \circ \psi^{-1} = (f \circ \varphi^{-1}) \circ (\varphi \circ \psi^{-1})$ is smooth at $\psi(p)$.

Proposition 1.26. *Let M be a manifold of dimension n, and $f : M \to \mathbb{R}$ be a function. Then the following are equivalent:*

1. *f is a smooth function on M.*
2. *M has an atlas $\mathcal{U} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$ such that $f \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha) \to \mathbb{R}$ is smooth for all $\alpha \in \Lambda$.*
3. *For all charts (U, φ) on M, the map $f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}$ is smooth.*

Proof. Note that 2. implies 1. by definition, and 3. implies 2. trivially. We show that 1. implies 3. Let (U, φ) be a chart on M, and $p \in U$; it is enough to prove that $f \circ \varphi^{-1}$ is smooth at $\varphi(p)$. Since f is smooth, there exists a chart (V, ψ) of M such that $p \in V$ and $f \circ \psi^{-1} : \psi(V) \to \mathbb{R}$ is smooth at $\psi(p)$. Now, we have

$$f \circ \varphi^{-1} = (f \circ \psi^{-1}) \circ (\psi \circ \varphi^{-1}). \tag{1.61}$$

The right-hand side is a composition of smooth maps, and hence is smooth at $\varphi(p)$. Thus, $f \circ \varphi^{-1}$ is smooth at $\varphi(p)$. ∎

A function $f : M \to \mathbb{R}$ is smooth on a chart (U, φ) if and only if the pullback $\varphi^* f := f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}$ is smooth. Also note that the definition of a smooth function f does not assume that f is continuous; however, one may show that the definition itself implies that f is continuous. This is because if f is smooth at $p \in M$, then there exists a chart (U, φ) of M such that $f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}$ is smooth at $\varphi(p)$, and hence continuous at $\varphi(p)$. Since φ is a homeomorphism, we have that f is continuous at p. Since p was arbitrary, we have that f is continuous on M.

Definition 1.27. Let M and N be smooth manifolds of dimension m and n, respectively. A continuous map $F : N \to M$ is smooth at $p \in N$ if there exist charts (U, φ) on N, with $p \in U$, and (V, ψ) on M, with $F(p) \in V$, such that $\psi \circ F \circ \varphi^{-1} : \varphi(U) \to \psi(V)$ is smooth at $\varphi(p)$. A continuous map $F : N \to M$ is smooth if it is smooth at all $p \in N$.

The definition of a map f being smooth at p is independent of the choice of charts (U, φ) and (V, ψ), since if (U', φ') and (V', ψ') are other charts containing p and $F(p)$, respectively, then $\psi' \circ \psi^{-1} : \psi(V \cap V') \to \psi'(V \cap V')$ and $\varphi' \circ \varphi^{-1} : \varphi(U \cap U') \to \varphi'(U \cap U')$ are smooth, and hence

$$\psi' \circ F \circ \varphi'^{-1} = (\psi' \circ \psi^{-1}) \circ (\psi \circ F \circ \varphi^{-1}) \circ (\varphi \circ \varphi'^{-1}) \tag{1.62}$$

Proposition 1.28. *Let $F : N \to M$ be a continuous map between smooth manifolds. Then the following are equivalent:*

1. *F is smooth.*
2. *There exist atlases $\mathcal{U}$ of N and $\mathcal{V}$ of M such that for all $(U, \varphi) \in \mathcal{U}$ and $(V, \psi) \in \mathcal{V}$, the map $\psi \circ F \circ \varphi^{-1} : \varphi(U \cap F^{-1}(V)) \to \psi(V)$ is smooth.*
3. *For all charts (U, φ) on N and (V, ψ) on M, the map $\psi \circ F \circ \varphi^{-1} : \varphi(U \cap F^{-1}(V)) \to \psi(V)$ is smooth.*

Proof. Again, note that 2. implies 1. by definition, and 3. implies 2. trivially. A similar argument as in the previous proposition shows that 1. implies 3. ■

Proposition 1.29. *Let $F : N \to M$ and $G : M \to P$ be smooth maps between smooth manifolds. Then the composition $G \circ F : N \to P$ is smooth.*

Proof. Let (U, φ), (V, ψ), and (W, σ) be charts on N, M, and P, respectively, such that $p \in U$, $F(p) \in V$, and $G(F(p)) \in W$. Then we have

$$\sigma \circ (G \circ F) \circ \varphi^{-1} = (\sigma \circ G \circ \psi^{-1}) \circ (\psi \circ F \circ \varphi^{-1}). \tag{1.63}$$

The right-hand side is a composition of smooth maps, and hence is smooth at $\varphi(p)$. Thus, $G \circ F$ is smooth at p. ■

Finally, a smooth map $F : N \to M$ is termed a *diffeomorphism* if F is a bijection, and F and F^{-1} are smooth. Some properties of diffeomorphisms may be inferred.

Proposition 1.30. *The following hold true.*

1. *Let (U, φ) be a chart on a smooth manifold M of dimension n. Then $\varphi : U \to \varphi(U)$ is a diffeomorphism.*

2. *Suppose $U \subseteq M$ is open, and that $F : U \to F(U)$ is a diffeomorphism. Then (U, F) is a chart in the maximal atlas of M.*

Proof.

1. Note that U is a manifold since it is open in M, and $\varphi(U)$ is a manifold since it is open in $\mathbb{R}^n$. The map $\varphi : U \to \varphi(U)$ is a homeomorphism by definition of a chart, so it is bijective. Note that $\{(U, \varphi)\}$ is an atlas on U, and $\{(\varphi(U), \mathrm{id}_{\varphi(U)})\}$ is an atlas on $\varphi(U)$. Then φ is smooth since $\mathrm{id}_{\varphi(U)} \circ \varphi \circ \varphi^{-1} = \mathrm{id}_{\varphi(U)}$ is smooth, and φ^{-1} is smooth since $\varphi \circ \varphi^{-1} \circ \mathrm{id}_{\varphi(U)} = \mathrm{id}_{\varphi(U)}$ is smooth. Thus, φ is a diffeomorphism.

2. Since U is open in M, it is a smooth manifold. Because $F : U \to F(U)$ is a diffeomorphism, it is in particular a homeomorphism. Hence F is bijective, and its image $F(U)$ is an open subset of $\mathbb{R}^n$ (as required for a chart). Let (V, ψ) be any chart on M with $U \cap V \neq \varnothing$. Then the transition maps

$$\psi \circ F^{-1} : F(U \cap V) \to \psi(V) \quad \text{and} \quad F \circ \psi^{-1} : \psi(U \cap V) \to F(U \cap V) \tag{1.64}$$

are smooth, since they are compositions of smooth maps. Therefore (U, F) is compatible with every chart in the maximal atlas of M, and so (U, F) is a chart in that maximal atlas.

■

Proposition 1.31. *Let N be a smooth manifold of dimension n, and $F : N \to \mathbb{R}^m$ be a continuous map. Then the following are equivalent.*

1. *F is smooth.*

2. *N has an atlas $\mathcal{U}$ such that for all $(U, \varphi) \in \mathcal{U}$, the map $F \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}^m$ is smooth.*

3. *For all charts (U, φ) on N, the map $F \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}^m$ is smooth.*

Proof. 3. implies 2. is obvious. For 2. implies 1., note that there exist atlases $\mathcal{U}$ and $\mathcal{V}$ of N and M respectively such that $(U, \varphi) \in \mathcal{U}$ and $(V, \psi) \in \mathcal{V}$ satisfy the smoothness condition. Since $\mathbb{R}^m$ has a global chart $(\mathbb{R}^m, \mathrm{id}_{\mathbb{R}^m})$, we can take $\mathcal{V} = \{(\mathbb{R}^m, \mathrm{id}_{\mathbb{R}^m})\}$. Then for any $(U, \varphi) \in \mathcal{U}$, we have $\mathrm{id}_{\mathbb{R}^m} \circ F \circ \varphi^{-1} = F \circ \varphi^{-1}$ is smooth, which implies that F is smooth. For 1. implies 3., F is smooth and φ is a diffeomorphism; thus, φ^{-1} is smooth, and $F \circ \varphi^{-1}$ is smooth. ■

Proposition 1.32. *Let $F : N \to M$ be a continuous map between smooth manifolds. Then the following are equivalent.*

1. *F is smooth.*

2. *M has an atlas $\mathcal{V}$ such that for all $(V, \psi) \in \mathcal{V}$, the map $\psi \circ F : F^{-1}(V) \to \psi(V)$ is smooth.*
3. *For any chart (V, ψ) of M, the map $\psi \circ F : F^{-1}(V) \to \psi(V)$ is smooth.*

The proof follows a similar structure to the one above.

Proposition 1.33. *Let $F : N \to \mathbb{R}^m$ be a continuous map. Then the following are equivalent.*

1. *F is smooth.*
2. *$F^1, \ldots, F^m : N \to \mathbb{R}$ are smooth, where $F = (F^1, \ldots, F^m)$.*

Index